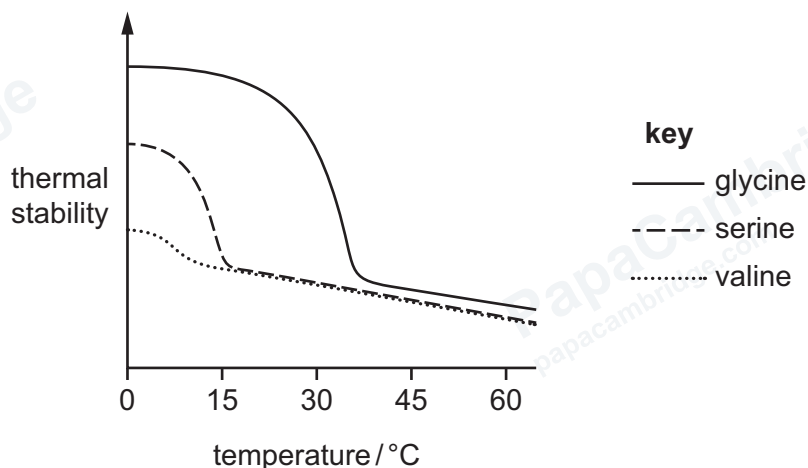


- 12 The thermal stability of collagen is measured by determining the temperature at which the three polypeptide subunits of collagen molecules become fully separated.

The graph shows how changing the amino acid sequence in the polypeptides that make up collagen molecules causes a change in the thermal stability of collagen.

Glycine is the amino acid with the smallest variable group. Scientists changed this amino acid in collagen polypeptides to either serine or valine.



A student analysed the graph and suggested three possible explanations.

- The variable groups of valine and serine are too large to allow close packing of collagen fibres.
- There are more hydrogen bonds present between collagen polypeptides that contain serine instead of glycine, than between collagen polypeptides that contain valine instead of glycine.
- The triple helices that make up collagen are less separated at higher temperatures in the protein with glycine than in the proteins with either serine or valine.

How many of these explanations could be correct?

- A 0      B 1      C 2      D 3